MATHS MANTHAN
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e-mail: contact@mathsmanthan.co.in | Helpline: +91-8171546558

Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 09** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 11: Areas Related to Circles** **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS (LA) questions of 5 marks each.
6. **Section E** comprises 1 Applied Case-Based question of 4 marks with internal choices.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. The wheels of a bus make 210 revolutions per minute. If the diameter of each wheel is 1 m, the speed of the bus in km/h is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 39.6 km/h (B) 19.8 km/h (C) 66 km/h (D) 33 km/h

Q2. Three sectors of a circle of radius r have central angles $35°, 65°$, and $80°$. The sum of their areas equals: [1]

(A) $\frac{1}{4}\pi r^2$ (B) $\frac{1}{3}\pi r^2$ (C) $\frac{1}{2}\pi r^2$ (D) $\frac{2}{3}\pi r^2$

Q3. From each vertex of an equilateral triangle of side 12 cm, a sector of radius 6 cm is drawn. The sum of the areas of the three sectors is: (Use $\pi = 3.14$) [1]

(A) 56.52 cm^2 (B) 113.04 cm^2 (C) 37.68 cm^2 (D) 18.84 cm^2

Q4. A right triangle has legs 6 cm and 8 cm. The area of its incircle is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 12.57 cm^2 (B) 25.14 cm^2 (C) 4π cm^2 (D) 16π cm^2

Q5. If a sector of central angle $60°$ has an area of $\frac{132}{7}$ cm^2, then the radius of the circle is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 6 cm (B) 7 cm (C) 5 cm (D) 8 cm

Q6. **Assertion (A):** A car wheel covers a distance of 22 km in 5000 complete revolutions if its radius is 70 cm.
Reason (R): The distance covered by a wheel in n revolutions is given by Distance $= n \times 2\pi r$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. Two concentric circles have radii 14 cm and 7 cm. Find the area of the circular ring bounded by them. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

Q8. The perimeter of a sector of a circle of radius 5.2 cm is 16.4 cm. Find the area of the sector. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. Calculate the area of the shaded leaf design between two quadrants of circles of radius 8 cm each drawn from opposite vertices of a square of side 8 cm. (Use $\pi = 3.14$) [3]

Q10. In a circle of radius 28 cm, an arc subtends an angle of 45° at the centre. Find the length of the arc and the area of the sector. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [3]

SECTION D: Long Answer / HOTS Type (2 × 5 = 10 Marks)

Q11. In a circle of radius 12 cm, a chord AB subtends an angle of 120° at the centre O.

(a) Draw altitude $OM \perp AB$ and derive the exact formula for the area of $\triangle OAB$ in terms of radius r and angle 120°.

(b) Calculate the area of the minor segment bounded by chord AB.

(c) Calculate the area of the corresponding major segment. (Use $\pi = 3.14$, $\sqrt{3} = 1.73$)

[5]

Q12. A designer creates nine identical circular floral motifs of radius 7 cm on a square silk handkerchief of side 42 cm.

(a) Find the total circumference of the nine circular motifs.

(b) Find the total area occupied by the nine circular motifs.

(c) Find the area of the remaining silk handkerchief.

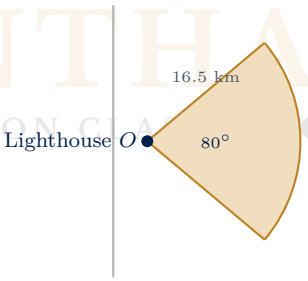
(d) If the remaining portion is dyed at the rate of 0.15 per cm^2, find the total dyeing cost. $\left(\text{Use } \pi = \frac{22}{7}\right)$

[5]

SECTION E: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q13. Maritime Lighthouse Harbor Warning Sector

A coastal port authority maintains a powerful lighthouse situated on a rocky promontory. The lighthouse beacon projects an intense red warning beam through a sector angle of 80° across the sea surface to a maximum distance of 16.5 km to steer vessels away from shallow reefs.



(i) State the geometric relationship between the sector radius and the light range. [1]

(ii) Calculate the length of the sea boundary arc illuminated by the warning beam. [1]

(iii) Calculate the total area of the sea over which vessels receive navigation warning, OR if the light range is increased to 20 km, find the new warning coverage area. (Use $\pi = 3.14$) [2]

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